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Moisture Conservation Practices

for enhancing productivity in rainfed agriculture

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INDIAN agriculture virtually depends on the arrival of the monsoon and small farmers rely on rain to irrigate their fields. Dryland and rainfed agriculture has a distinct place in Indian agriculture, occupying around 108 m ha area (75%) out of 143 m ha net cultivated area. Despite considerable progress in irrigation development since 1950s, 85.6% of coarse cereals, 83.8% of pulses, 42% of rice, 74.1% of oilseeds and 64.7% of cotton are still cultivated as rainfed. Dryland and rainfed agriculture contributes 40% of foodgrains production and supports half of the human and two-third of the livestock population. This implies that the country will continue to grapple with the problems of rainfed agriculture. Dryland farming is characterized by low crop productivity and high variation in yields from year to year. In dry farming areas, though the total rainfall received is adequate for crop growth, its distribution determines the crop productivity. The dry spells, due to erratic rainfall distribution, especially at critical stages, affect the yield adversely. In rainfed areas, all the rainfall received is not available for the crops and a significant part is lost as run off. Moisture stress further affect the nutrient availability to the crop since nutrient mobility depends on optimum soil moisture. In fact, successful dry farming depends chiefly upon the success with which the rains that fall during any season of the year may be stored and kept in the soil until needed by plant for their growth. The fundamental operation of dryland farming includes a soil treatment which enables the largest possible proportion of the annual precipitation to be stored in the soil.

India will have to produce 300 million tonnes of food grains to feed her burgeoning population. This target cannot be realized from irrigated areas alone. An appropriate technology has to be evolved for dry land farming. On the other hand, it can be said that second 'green revolution' in Indian agriculture can be had in rainfed/dryland agriculture. This is important to improve the standard of living of farmers residing in these areas as well.

Problems of dryland and rainfed areas

Most of the cropping in the arid and semi-arid regions continues to be under rainfed conditions. Dryland soils are not only thirsty but hungry also. A majority of the farmers are small farmers with meagre resources. The poor resource base permits only low input subsistence farming with low and unstable crop yields. The low productivity of agriculture in dryland farming regions is due to the cumulative effect of many constraints for crop production. Frequent drought, little or no biomass recycling and low soil organic matter (SOM), multi-nutrient deficiencies, depletion of nutrients by crops, sub-optimal nutrient application, low nutrient use efficiency *etc.* are the major constraints in enhancing productivity in the rainfed and dryland areas. Because of the following reasons at present, 3 ha of dryland crop produce cereal grain equivalent to that produced in one ha irrigated crop. The main problem of rainfed areas is uncertainty and uneven distribution of rainfall and loss of water through runoff which leads to low and unstable productivity due to moisture stress at critical stages of crop growth. It is well known fact that about 85 per cent of annual rainfall is received during south-west monsoon season. In this period knowledge of crop growth phases and moisture availability is more essential because the deficiency of rain water at any critical growth stage may affect the plant growth and yield. Moisture stress further affect the nutrient availability to the crop since nutrient mobility depends on optimum soil moisture. There the major burden in

India has about 108 million hectares of rainfed area which constitutes nearly 75% of the total 143 million hectares of arable land. In such areas crop production becomes relatively difficult as it mainly depends upon intensity and frequency of rainfall. The crop production, therefore, in such areas is called rainfed farming as there is no facility to give any irrigation, and even protective or life saving irrigation is not possible.

future food needs of the country depends on how best rainfed regions are managed and productivity levels are improved. Hence concerted efforts are needed to develop soil and moisture conservation practices to mitigate the water stress to maximize food production with minimum environmental degradation.

Moisture Conservation Practices for Rainfed and Dryland Areas

The productivity of grains already showed a plateau in irrigated agriculture due to problems related to nutrient exhaustion, salinity build up and raising water table. Therefore, the challenges of the present millennium would be to produce more from dryland while ensuring conservation of existing resources, the soil and water. Hence, new strategies would have to be used which would conserve the water and the fragile soil of dryland ecosystems.

Crop and varietal selection

***Choice of crops*:** In dryland areas the traditional cropping pattern is dominated by foodgrain crops viz. millets and pulses. The suitability of the crop in the region depends on many factors like rainfall pattern, onset of rainfall season, length of the rainy season, soil type and socio-economic conditions of the farmers. As the arid zone has been endowed with variety of soils the choice of crops should be based on nutrient and moisture requirement vis-à-vis soil type. The soil fertility is quite low for crop production considering the nutrient status of the soil and nutrient requirement of the crops. The major focus of research under AICRPDA has been on the identification of most efficient crops for each dryland farming region. The criteria for choice of crops for dryland farming regions comprise the following,

- Tolerance to drought.
- Fast growth during initial period to withstand harsh environment.
- Genetic potential for high yield.
- Short or medium duration to escape terminal drought.
- Adaptability to wide climatic variations.
- Responsive to fertilizers.

***Selection of suitable varieties*:** In dryland and rainfed areas farmers still are growing the traditional local varieties of most of the crops. The preference for these local varieties is based on their pronounced drought tolerance. But they are usually longer in duration susceptible to moisture stress at maturity. Choice of a suitable variety plays paramount role for obtaining efficient and successful crop production in rainfed areas. The criteria now adopted for selection of crop varieties for drylands include drought tolerance, short or medium duration, high yield potential, response to nutrient supply, high water use efficiency and moderate resistance to pest and diseases. Suitable varieties for all dryland crops have been developed in all the dry farming regions and have proved their high yield potential. Some improved varieties for rainfed areas are given in Table 1.

Land configuration and tillage management

Land configuration and tillage management are the practices of manipulating or managing the soil surface for the purpose to retain the maximum rain water in the soil profile while providing optimum conditions for growth of plant. This can be achieved by different measures. The different land configuration measures for *in-situ* moisture conservation practices which result in changed land configuration are as follows:

Ridge and furrow: This practice is suitable for wide row spaced crops like cotton, pearl millet, maize, chillies, tomato *etc.* Furrows of 30-45 cm width and 15-20 cm height are formed across the slope. The furrows collect and store water when rainfall intensity is less and guide runoff water safely when rainfall intensity is high and avoid water stagnation. It is suitable for medium deep to deep black soils and deep red soils. It is not suitable for shallow red soils, shallow black soils and sandy gravelly soils. It is not suitable for broadcast sown crops and for crops sown at closer row spacing less than 30 cm. Sowing on ridges and furrows recorded lower weed biomass and higher water use efficiency of cotton. It has also been reported that planting of pearl millet and mustard under ridge and furrow system recorded significantly higher grain yield, consumptive use and water use efficiency as compared to their flat sowing.

Broad bed furrows (BBF): It implies shaping of land into alternate beds and furrows. This practice has been

Table 1. Improved crop varieties suited to rainfed agro-ecosystems

Crop	Variety
Sorghum	CSH 5, CSH 6, CSH 9, CSV 8, CSV 9, CSV 10, CSV 15
Maize	Ageti 76, Prakash, Pusa composite 3, Shakti, Ratana
Pearl Millet	Pusa composite 443, HHB 60, HHB 67, Raj 171, Pusa 444, BK 560, RHB 30, RHB 90, RHB 121, PHB 10
Wheat	K 65, WH 410, VL 804, VL 738, VL 421,, VL 829, PBW 396, K 8027, HDR 77, Amar, HI 1500
Lentil	Pusa Vaibhav, Shivalik, Pant L 406, Pant L 639, DPL 15
Pigeonpea	Pusa Ageti, Sharda, Prabhat, UPAS 120, H 1, H 3, Gwalior 3
Blackgram	T 9, Krishna, IPU 94-1
Greengram	Pusa Baisakhi, RS 4, K 851, RMG 62, RMG 268, Pusa Vishal
Chickpea	Pusa 256, Pusa 1053, Pusa 1103, C 235, Pusa 372, Pusa 362
Clusterbean	Durgapura Safed, FS 277, Maru guar, HG 75, G 1, Suvidha
Taramira	T 27, RTM 314 (Karan Tara), RTM 1, RTM 2, RTM 2002
Sesame	TC 25, TC 46, T 13, N 32, PB Til, RT 124, RT 125, RT 142
Groundnut	TMV 1, M 13, T 64, S 206, TG 26, TG 30, K 134, ICGS 92234
Castor	Aruna, GCH 3, Jyoti, GCH 4, PCH 1, Bhagya, TMV 5, TMV 6
Safflower	JSF 10, A 300, N 62-8, N 7, Tara, NARI 6, Bhima, Sharda
Soybean	JS72-44, Ankur, Bragg
Sunflower	EC 68414, EC 68415m MSFH 17, KBSH 1, Moden
Indian mustard	RH 30, Pusa Bold, Aravali, Geeta, Pusa Bahar, Varuna, Kranti
Mothbean	Maru Moth 1, RMO 40, FMM 96, RCMO 912
Horse gram	Maru Kulth 1, Solapur, TPK 2, PHG 9
Barley	Ratna, DL 3, Azad, Ambar, RS 6, Kailash, Karan 795, Vijaya



Ridge and furrow method



Deed furrows

recommended by ICRISAT for vertisols or black soils in high rainfall areas (>750 mm). Here beds of 90-120 cm width, 15 cm height and convenient length are formed, separated by furrows of 60 cm width and 15 cm depth. When runoff occurs, its velocity will be reduced by beds and infiltration opportunity time is increased. The furrows have a gradient of 0.6%. Crops are sown on the broad beds and excess water is drained through number of small furrows which may be connected to farm ponds. It can be formed by bullock drawn or tractor drawn implements. Bed former-cum-seed drill enables BBF formation and sowing simultaneously, thus reduces the delay between receipt rainfall and sowing. Broad bed and furrow has many advantages over other methods. Planting of sunflower under BBF method registered higher soil moisture content and seed yield as compared to flat sowing.



Broad bed furrows

Dead furrows: Dead furrows (idle furrow) are laid across the land slope in rolling lands to intercept the run-off. At the time of sowing or immediately after sowing, deep furrows of 20 cm depth are formed at intervals of 2 to 5 m or 6 to 8 rows of crops. This system worked well with Alfisols. No crop is raised in the furrow. The dead furrow reduces velocity of runoff and increases the infiltration opportunity time.

Tie-ridging: The practice of tie-ridging, where adjacent ridge are joined at regular intervals by barriers or ties of the same height, allows the water to infiltrate and prevent runoff expect during intense storms. This method is adequate in moderate rainfall areas, except on very steep slopes.



Tie ridging

Compartmental bunding: This method is one of the cheapest rainwater conservation technique suitable for rainfed vertisols having slope of less than 1%. By this method the entire field is divided into small sections, which helps in storing the initial rainfall and permitting increased infiltration rate. This ensures rational moisture distribution as well as optimum moisture for better germination of seeds. In *kharif* fallowing, this practice enhances enough rain water inventory in the soil profile for raising *rabi* crops on the conserved moisture.



Compartmental bunding

Tillage: Tillage is a well-known soil and water conservation practice which makes soil surface more permeable to increase infiltration of rainwater into the soil, which in turn reduces runoff, soil and nutrient losses and enhance crop yields. Deep ploughing is taken up to break hard pan and it also improves physical characteristics of black soils. It has to be done to a depth of 22 cm (tractor plough) in Vertisols once in 3 years with faster penetration of rain water, eradicate pernicious weeds and results in higher crop yields. In dry farming, the soil becomes vulnerable for hardness when kept fallow during summer months. At the instance of rainfall, the raindrops beat the top surface of soil and make them split and cause heavy loss of top soil transported alongwith the runoff waters. To prevent loss of top soil and runoff, off-season or summer ploughing is done which helps the soil to absorb the rains during monsoons and conserve the same as soil moisture for crop growth. Conservation tillage, which ensures at least 30% coverage of the soil surface with crop residue play very important role in organic carbon build up and soil and moisture conservation under dryland. Conservation tillage is an umbrella term and it envisages stubble mulch tillage, minimum tillage and zero tillage. Low intensity tillage favours consolidation of soils and imparts erosion resistance through better consolidation, structure, infiltration, pores distribution and depression storage.

Soil conditioners: Soil conditioners are termed as materials which when added to the soil help in improving or maintaining its physical conditions with improved physical and chemical health of soil. Under different soil conditioners hydrophilic polymers are used to improve the moisture holding capacity of soil. Pusa Hydrogel is an indigenous product designed and developed by scientists of Indian Agricultural Research Institute, New Delhi to enhance the crop productivity per unit available water and nutrients, particularly in moisture stress agriculture. This gel exhibits maximum absorption capacity at temperature above 45^{0}C, absorbs water 350-500 times it's dry weight and gradually releases the same into the soil. Pusa Hydrogel remains stable in soil for a minimum period of one year and less affected by salts. Only a moderate volume of this gel, such as 2.5 to 5.0 kg is required to be added per hectare of land against as much as 10 kg of other gels used elsewhere in world. Presently the cost of manufacturing Pusa Hydrogel ranges between ₹ 1,000 to 1,400 a kg, but the commercial returns from

its deployment are as high as five to seven times the manufacturing cost. Other soil conditioner is Jalshakti (a hydrophilic polymer) which when applied (mixed) in soil, improves moisture holding capacity of soil, increase infiltration of water, reduce evaporation from soil and prolong the availability of soil moisture to the crop.

Mulching: Mulching may be defined as the process of covering the soil surface around the plants to create congenial conditions for the crop growth. In other words, mulching is the process of applying any material (organic or inorganic) on the soil surface to check evaporation and improve soil water. Mulches in addition to reducing evaporation losses of moisture also bring about conspicuous effect on numerous soil properties and conditions directly and indirectly, such as runoff control, increased infiltration, erosion control, regulation of soil temperature, improvement of soil structure, control of weeds and reduction in salinity. Mulches are of different types, as given further in the article.

Soil mulch or dust mulch: A soil management practice to improve water conservation, dust mulches are created by intensively hoeing the soil surface, creating a finely-textured layer of soil. Dust mulching breaks the soil capillarity, reducing the evaporative loss of soil moisture. Some scientists observed that a surface mulch of dry soil 5-8 cm deep by obstructing the rise of water to the surface through capillary action, effectively reduced loss of water as compared with a soil having an undisturbed surface.



Soil mulch

Stubble mulch: Stubble mulching is leaving the organic residue of the crop that is not gathered during harvest on the ground. Crop residues like wheat stubble or cotton stalk *etc.*



Stubble mulch

are left on the soil surface as stubble mulch. Stubble-mulching can prevent soil erosion from wind or water and conserve soil moisture.

Straw or stover mulch: This involves application of straw or stover to create a protective cover over the entire surface or a part of the soil surface for reducing evaporative losses of water and conservation of water. It also helps in reduction of soil erosion, soil temperature management and suppresses weed growth.



Straw or stover mulch

Live mulch: Despite the beneficial effects of mulching, adoption of this practice is constrained due to non-availability of mulch material. There is a possibility of biomass production and nutrient cycling through live mulching of *in-situ* grown annual legumes along with field crops. Live



Live mulch

mulching (mix of fast growing plants that are planted around the main crops) is an excellent way to rejuvenate arid land or fill areas which don't have plans to be cultivated. Live mulch with fast-growing ground cover crop also provides weed suppression and aids in nitrogen fixing.

Plastic mulch: When compared to other mulches plastic mulches are completely impermeable to water; it therefore prevents direct evaporation of moisture from the soil and thus limits the water losses and soil erosion over the surface. It plays a positive role in water conservation. The suppression of evaporation also has a supplementary effect; it prevents the rise of water containing salt, which is important in countries with high salt content water resources. Plastic materials like polyethylene, polyvinyl chloride of 25-50 micron thickness are used as mulching materials. It should be relatively cheaper, resistant to biochemical degradation, allow low water vapour permeability, good transparency and free from toxicity.



Plastic mulch

Vertical mulch: It is an excellent technique used in black soil where the infiltration of water is very slow and much of the rainfall is lost as runoff. To improve infiltration and storage of rain water in these soils, vertical mulches are used. It is a technique wherein trenches of 40 cm wide, 15 cm deep are dug at 2 to 4 m interval across slope and filled with stubbles or organic wastes to a height of 10 cm above soil surface. Runoff is checked, collected in the shallow trenches and redistributed to adjoining soil layers and infiltration is increased.



Vertical mulch

Transpiration suppressant

Any material applied to transpiring plant surfaces with the aim of reducing water loss from the plant is called transpiration

suppressant or anti-transpirant. The main objective to transpiration suppressant is that by reducing transpiration they might cause an increase in leaf temperature to critical levels. It has however, been estimated that under field conditions, transpiration cannot reduce the temperature of leaves by more than 5 % and that under moist conditions the higher temperature would not be critical. The best anti-transpirant reduce transpiration by 30 to 40 % without causing much effect on CO_2 diffusion and photosynthetic rate. There are 4 types of transpiration suppressants:

Stomatal closure or metabolic type: Some fungicides like PMA (phenyl mercuric acetate) and herbicides like atrazine in low concentrations serve as transpiration suppressant by closing of stomata. These might also reduce the photosynthesis simultaneously. PMA is known to reduce transpiration to a greater degree than photosynthesis in several plants. Various herbicides like atrazine, and other chemicals viz. alachlor, abscisic acid (ABA) etc. when sprayed on crop foliage in low concentration were found to act as transpiration suppressants. These spray materials have been found to cause the closure of the guard cells of the stomata and thereby reduce transpiration.

Film forming types: Silicon's, polyethylenes, hexadecanol, mobileaf, wax, latex, fatty alcohols etc. are some of the film forming type of transpiration suppressants. The success of these materials is limited because they also reduce diffusion of CO_2 from leaf and ultimately reduce the rate of photosynthesis. A good film forming type of transpiration suppressant is that which should form a thin layer, that should be more resistant to the passage of water vapour than CO_2 and the film should maintain continuity for a longer period of time.

Reflectant type: Reflectant types of transpiration suppressants are the suspension of material in water which when sprayed on the foliage, a thin white coating is formed that increase leaf reflectance (albedo) and decreases the net radiation load and temperature which results in the reduction of vapour pressure gradient and eventually the transpiration. White material like kaolin, lime, hydrated lime, zinc sulphate, magnesium carbonate etc. could be used as light reflectant. It has been investigated that application of 6% kaolin in combination with organic mulch control, transpiration and enhance the consumptive use and water use efficiency with significant increase in grain yield.

Growth retardant type: Growth retardants reduce the total surface area by reducing the shoot growth and by increasing root growth and thus enable the plants to reduce transpiring surface and resist drought conditions. They increase root/shoot ratio. Cycocel or CCC (2-chloroethyl, trimethyl ammonium chloride) is one such chemical useful for improving water status of the plant. Application of cycocel at 200 ppm concentration is helpful in reducing the water loss through transpiration and also enhances the moisture use efficiency and consumptive use of water.

Ex-situ water conservation

Rainwater runoff varies from 10-40% of the total precipitation depending upon the rainfall intensity and distribution, soil characteristics (slope, water intake rate and water holding capacity) and vegetative cover. *Ex-situ* water conservation is the process of collecting and concentrating runoff water during periods of peak rainfall into an on-farm water reservoir such small farm pond, tank etc. for future productive uses. The excess rainwater is lost as runoff causing soil erosion. Low moisture retention capacity of sandy soils further deteriorates the situation. Therefore, rainwater harvesting will not only conserve the soil, its fertility and vegetation; but also could be utilized as supplementary irrigation that will be advantageous in enhancing total water supply available to crop plants during low rainfall period. Harvested water is used to provide protective/ supplemental irrigation for survival of crops in drought prone areas with erratic rainfall. Critical irrigation with a small quantity of water harvested in a pond is of immense value to improve and stabilize the dryland crop production and to build up confidence in farmers of dryland areas. Such irrigation when provided at critical growth stage or as pre-sowing irrigation can increase crop yield many fold.



Farm pond



Percolation pond



Check dams

SUMMARY

Water is the key input for crop production in rainfed and dryland areas, therefore there is need to adopt a comprehensive approach to conserve soil moisture. Selection of suitable crops and their varieties is most important for rainfed and dryland areas. Planting of crops by altering land configuration viz. ridges and furrows, BBF, FIRB and bed planting are better for higher water productivity in both rainfed and dryland areas.

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